Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
void fun(int n){
    int counter = 1;
    if(n == 1){
        printf("*"); counter += 2;
    }
    else{
        for(int i=1; i<=n; i++){
           counter++;
            for(int j=1; j<=n; j++){
                counter++;
               // printf("*");
               // printf("*");
               counter += 3;
               break;
            }
        }
        counter++;
       // counter += n+1; //Loop condition and False check
       // counter += n*2; //Inner loop runs n-1 times and 2 print statements
    }
    printf("%d",counter);
}
int main(){
    int n;
    scanf("%d",&n);
    fun(n);
    return 0;
```

```
29 }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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